

More Evidence on the Unergative-Unaccusative Distinction in Second Language Grammars

Takayuki Kimura¹ and Takaaki Hirokawa²

¹*Utsunomiya University*

²*University of Fukui*

Abstract

This study presents new evidence for the structural unergative-unaccusative distinction, in second language (L2) grammars, focusing on elementary-level Japanese-speaking learners of English (JLEs). The underlying distinction of unergatives-unaccusatives is often obscured on the surface strings due to independent syntactic properties such as feature-driven subject movement (in English) or headedness (in Japanese). Nevertheless, based on previous findings, elementary-level JLEs are expected to have reset headedness but have not acquired subject movement. Then, the resulting representation would not involve the properties obscuring the underlying unergative-unaccusative distinction and potentially exhibit it on the surface strings in L2 English. Following these observations, we carefully designed test sentences with un/grammatical word orders that elementary-level JLEs would generate and conducted an acceptability judgment task with native speakers of English and elementary-/intermediate-level JLEs. The results showed that, in contrast to native controls and intermediate learners, who exhibited target-like patterns, elementary-level JLEs incorrectly accepted ungrammatical word orders only with unaccusatives (e.g., **When did arrive the train?*) but not those with unergatives (e.g., **Where did dance the man?*). This discrepancy can be attributed to the sensitivity to the structural distinction of unergative-unaccusative verbs, and our data provide evidence for the creative construction of an interlanguage in L2 acquisition.

1. Introduction

One of the key topics in second language (L2) acquisition of grammar is *argument structure*, which describes the configuration of arguments associated with verbs.¹ Verbs

¹ Generative L2 research has increasingly focused on feature reconfiguration since Chomsky (2001), significantly advancing our understanding of L2 phenomena (see, e.g., Lardiere, 2009; Lohndal & Putnam, 2024; Leal & Shimanskaya, 2024). However, as we will argue, a fundamental research question regarding argument structure remains unresolved.

may require arguments, but their syntactic positions differ by verb type. Intransitive verbs, selecting one argument, fall into two categories: Unergative verbs require an external argument (EA) at Spec-vP (1a), while unaccusative verbs select an internal argument (IA) at Comp-VP (1b) (Burzio, 1986; Perlmutter, 1978):

- (1) a. [_{VP} **DP** [_{VP} V ____]]
 b. [_{VP} ____ [_{VP} V **DP**]]

The structural difference is reflected in the θ -roles that DPs receive. DPs at Spec-vP are (typically) interpreted as an AGENT, and those at Comp-VP as a THEME of the predicate (Burzio, 1986; Perlmutter, 1978; cf. Baker, 1988; see also Progovac, 2015).² Consider the structures in (2).

- (2) a. [_{VP} the boy_{AGENT} [_{VP} run ____]]
 b. [_{VP} ____ [_{VP} happen the accident_{THEME}]]

In (2a), *the boy* is interpreted as an AGENT, while *the accident* in (2b) is a THEME. In natural language, the unergative-unaccusative distinction and θ -role assignment are considered part of Universal Grammar (UG). Consequently, researchers have investigated whether this property guides grammar construction in L2 acquisition (see e.g., Hirakawa, 2003; Zobl, 1989).

The primary aim of this study is to provide new evidence that L2ers are sensitive to the structural unergative-unaccusative distinction from an early stage. Since the ‘syntactic’ evidence for the unergative-unaccusative distinction in L2 is difficult to observe, we carefully constructed specific possible erroneous sentences that L2ers (in this study, Japanese-speaking learners of English) would likely generate, based on insights from previous studies. Through testing learners’ knowledge about these sentences, we will argue for the underlying syntactic structures associated with intransitive verbs (unaccusatives vs. unergatives) in L2 grammar.

This paper is organized as follows: Section 2 reviews previous findings on L2 unaccusativity. Section 3 provides background on the syntax of subjects and unaccusativity in English and Japanese. Section 4, reviewing relevant literature, argues

² As an anonymous reviewer noted, Baker’s (1988) approach to θ -role assignment may be flawed. However, Chomsky (2021) recently suggested that θ -theory is integral to UG, and some fundamental principle must underpin θ -role assignment, regardless of the specific mechanism (see Kratzer, 1996; Progovac, 2015).

that Japanese-English interlanguage can provide evidence for deep unaccusativity. Section 5 details the experimental study, and Section 6 presents the results. Section 7 discusses the findings and their implications. Finally, Section 8 concludes the paper.

2. Unergative–Unaccusative Distinctions in L2 English

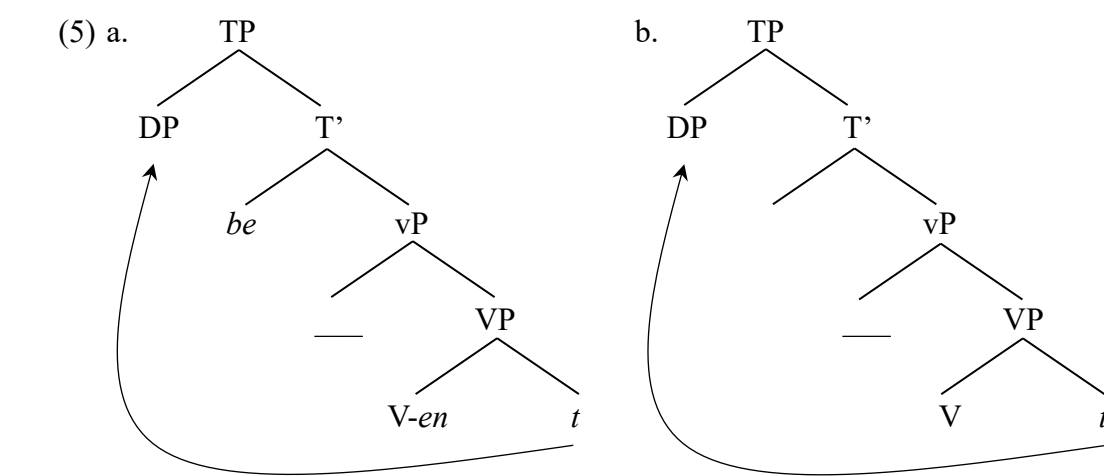
One of the most important findings in the study of L2 unaccusativity is the existence of ‘passive’ unaccusative errors found cross-linguistically, as shown in (3) (see e.g., Balcom, 1997; Ju, 2000; Hirakawa, 2003, 2006; Oshita, 2000; Zobl, 1989):

(3) *The accident *was* happened.

By contrast, ‘passive’ unergative errors are seldom found:

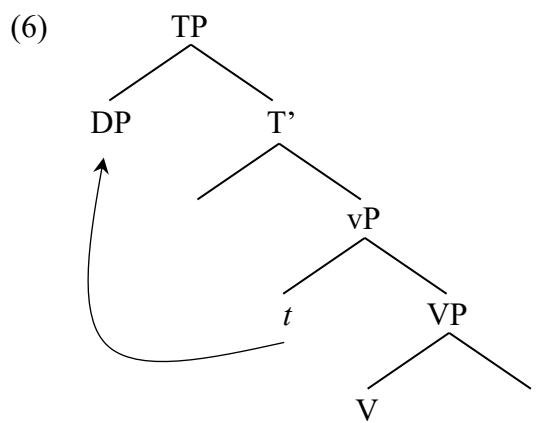
(4) *The boy *was* danced.

The syntactic explanation for the presence of ‘passive’ unaccusative errors and the absence of ‘passive’ unergative errors lies in the structural similarity between passive and unaccusative constructions.³ L2ers interpret *be* as signaling DP-movement to subject position (Zobl, 1989; Hirakawa, 2003, 2006; Oshita, 2000). Both passive and unaccusative constructions involve an IA without an EA. Since passivized verbs and unaccusative verbs lack a Case feature, the IA must move to Spec-TP to gain the Case value (5a,b) (Chomsky, 2001; Epstein, Kitahara & Seely, 2016; Jaeggli, 1986).



³ An anonymous reviewer pointed out that a finer analysis of passives (e.g., Alexiadou et al., 2015; Collins, 2005) could lead to different predictions or explanations. This may serve as a venue for further exploration.

Thus, the structural similarity between passives and unaccusatives may lead L2ers to overgeneralize the morphosyntactic properties of passives to unaccusatives. In contrast, unergatives, as shown in (6), involve the EA moving to the subject position, lacking the key properties of unaccusatives/passives. Therefore, overpassivization is not expected, consistent with previous findings.



Thus, the presence of ‘passive’ unaccusative errors and the absence of ‘passive’ unergative errors suggest the sensitivity to the unergative-unaccusative distinction.

Another piece of evidence for the unergative-unaccusative distinction comes from restrictions on secondary predication constructions, such as resultatives. In English resultatives (7), the resultative phrase *blue* is predicated of the immediately postverbal DP *the wall* (Levin & Rappaport-Hovav, 1995; see also Hoekstra, 1988; Carrier & Randall, 1992), meaning the wall became blue as a result of Jim’s painting.

(7) Jim painted [_{SC} the wall *blue*].

Resultatives follow the Direct Object Restriction (DOR), requiring the resultative phrase to be predicated of the postverbal DP (Levin & Rappaport-Hovav, 1995). The unergative-unaccusative distinction is evident in DOR’s applicability. For instance, the unaccusative verb *freeze* is compatible with *solid* (8a), while the unergative verb *shout* cannot have a resultative reading of a potential secondary predicate *hoarse* (8b).

- (8) a. The river froze solid.
 b. *Dora shouted hoarse.

As shown below, (8a) meets the DOR because unaccusative verbs select an IA (e.g., *river*

in (9a)) in the underlying structure. In contrast, (8b) fails to satisfy the DOR since unergatives lack an IA (9b).

- (9) a. [_{VP} ____ [_{VP} froze the river solid]]
 b. *[_{VP} Dora [_{VP} shouted ____ hoarse]]

Park and Lakshmanan (2007) examined whether Korean learners of English at intermediate and advanced levels are sensitive to the DOR. Their findings showed that native English speakers differentiated verb types, and L2 groups did as well, though L2ers were less sensitive to (un)grammaticality. The fact that L2ers distinguished between unergative and unaccusative verbs supports the presence of this distinction in L2 grammar.

Despite these interesting pieces of evidence, challenges remain. First, several researchers have argued against a syntactic account of ‘passive’ unaccusative errors (see, e.g., Yusa, 2003). Second, the DOR is not universally applicable (10), leaving room for reconsideration of its validity.

- (10) a. Robin danced ____ out of the room.
 (Rappaport-Hovav & Levin, 2001:782)
 b. The wise men followed the star out of Bethlehem.
 (Wechsler, 1997:313)

These examples suggest that the difference in (8a/b) may not stem from the unergative-unaccusative distinction.⁴ Therefore, more robust evidence for the unergative-unaccusative distinction in L2 grammar is necessary, and this study aims to contribute further evidence from a new perspective.

3. Syntax of Subjects and Unaccusativity in Japanese

This study examines the grammar of Japanese-speaking learners of English (JLEs) for reasons explained below. We first review the background of Japanese syntax. Notably, in Japanese, the distinction between unergative and unaccusative VP constructions is obscured in surface word orders:

⁴ We would like to thank an anonymous reviewer for pointing out this issue.

- (11) a. Syonen-ga warau _____. *unergative*
 boy-NOM laugh
 'A boy laughs.'
- b. ____ mado-ga wareta. *unaccusative*
 window-NOM broke
 'A window broke.'

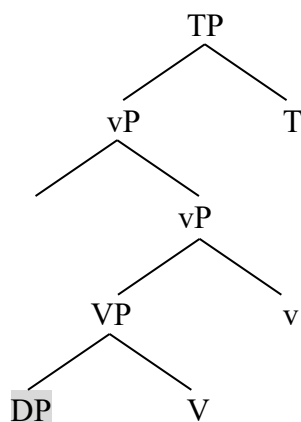
However, various researchers have found evidence for deep unaccusativity in Japanese (Kishimoto, 1996; Hirakawa, 2001). To give an example, *kake-no*-NP 'be-about-to-NP' can co-occur with unaccusative verbs but not with unergatives:

- (12) a. kare-kake-no hana
 wither-KAKE-GEN flower
 'a flower, almost withered'
- b. *sakebi-kake-no kankyaku
 shout-KAKE-GEN spectator
 'spectators, who are about to shout'

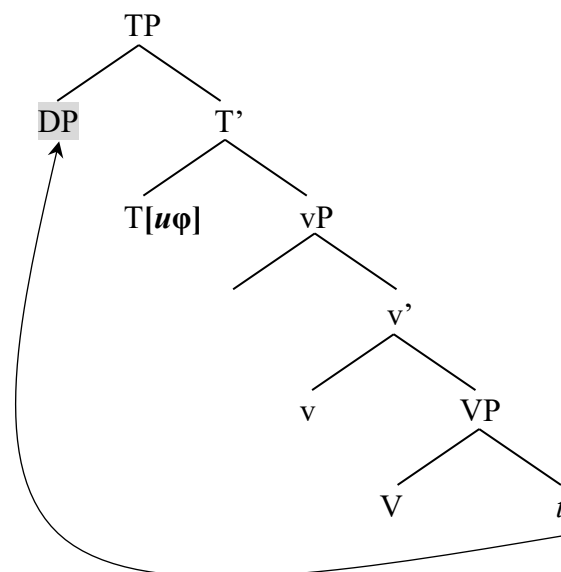
(Kishimoto, 1996: 257)

Therefore, the apparent lack of unaccusativity must be attributed to other factors. Notably, subjects remain in situ within vP in Japanese in the canonical word order (e.g., Fukui, 1995; Yatsushiro, 1999), unlike in English, where they obligatorily move to the subject position (Spec-TP), due to the presence of an uninterpretable ϕ -feature (hereafter, $u\phi$ -feature, Chomsky, 2001) as shown below:

(13) a. Japanese



b. English



Additionally, Japanese is a head-final language where V appears to the right of other elements within VP, as shown in (13a). For unergatives, DP_{AGENT} remains at Spec-vP, resulting in the [DP_{AGENT}>V] order (14a). For unaccusatives, an DP_{THEME} and a verb appear within VP, with the DP staying there, resulting in the [DP_{THEME}>V] order (14b), which obscures the surface unergative-unaccusative distinction.

- (14) a. [_{VP} Syonen-ga ‘boy-NOM’ [_{VP} warau ‘laugh’ ____]] *unergative*
 b. [_{VP} ____ [_{VP} Mado-ga ‘window-NOM’ wareta ‘broke’]] *unaccusative*

In summary, ‘deep’ unaccusativity is observed in Japanese, but the unergative-unaccusative distinction is lost on the surface due to head-finalness and the absence of *uφ*-driven/obligatory subject movement.

4. Japanese-English Interlanguage as Potential Evidence for Underlying Structures

Given above, if the L2er fails to acquire obligatory *uφ*-driven subject movement and has acquired the [head-initial] setting, they would show the unergative/unaccusative distinction on the surface. Based on previous findings, non-proficient JLEs would develop such an interlanguage.

Let us examine this in detail. First, regarding head-direction, it is well-established that the head-direction parameter is reset correctly very early in L2 development (e.g., Haznedar, 1997), with even early-stage learners rarely making errors in this area. Thus, elementary-level JLEs are expected to have reset the head-direction parameter from [head-final] (OV—Japanese value) to [head-initial] (VO—English value).⁵

Regarding obligatory subject movement, previous studies suggest that non-advanced learners are unlikely to acquire it (e.g., Kimura, 2022b). This prediction is supported by several assumptions.⁶ Assuming L1 Transfer (cf. Schwartz & Sprouse, 1996), JLEs would place subjects in situ in L2 English as they do in L1 Japanese. Additionally, since DPs in Japanese can optionally be fronted to the subject position via a non-feature-driven movement operation, *scrambling* (cf. Saito, 2003), both configurations are expected in their L2 English:

⁵ Note that our elementary JLEs have been exposed to L2 input for over 10 years.

⁶ This assumption seems reasonable given the following argument, though further empirical justification is desired. However, obtaining independent evidence for subject movement is challenging, and we leave it for future research. Setting it aside, the results discussed later support this assumption, as learners accepted sentences without subject displacement.

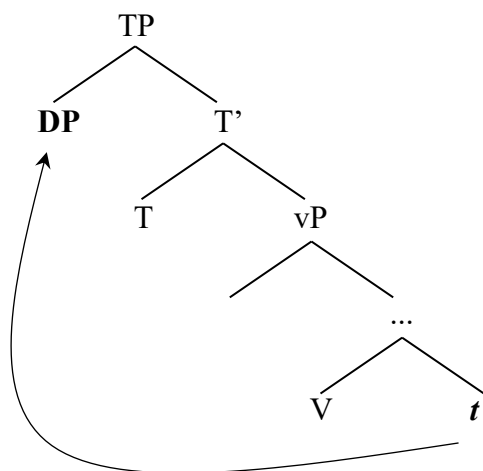
- (15) a. [TP____[_{VP} **DP** [_{VP}]] *without scrambling*
 b. [TP **DP** [_{VP}____[_{VP}]] *with scrambling*

As noted, obligatory subject movement in English is triggered by $u\phi$ -features (Chomsky, 2001), which JLEs must newly acquire. However, acquiring an uninterpretable feature absent in the learner’s L1 is notably difficult, even for advanced learners, as confirmed across different languages and constructions (e.g., Hawkins & Liszka, 2003; Hawkins & Hattori, 2006; Tsimpli & Dimitrakopoulou, 2007). Indeed, it is argued that acquiring subject movement in English is challenging for JLEs (e.g., Nakayama & Yoshimura, 2019; Kimura, 2022b). Additionally, Miyamoto and Iijima (2003) found that elementary JLEs use optional scrambling (sometimes in-situ, sometimes fronted) for English *wh*-questions, which require obligatory *wh*-movement triggered by an uninterpretable feature absent in Japanese.⁷ Collectively, it is highly likely that elementary-level JLEs fail to acquire the ϕ -feature and will employ optional scrambling for subject movement.⁸ This analysis allows for two alternative derivations in (16a,b, instances of unaccusative cases) in their interlanguage:

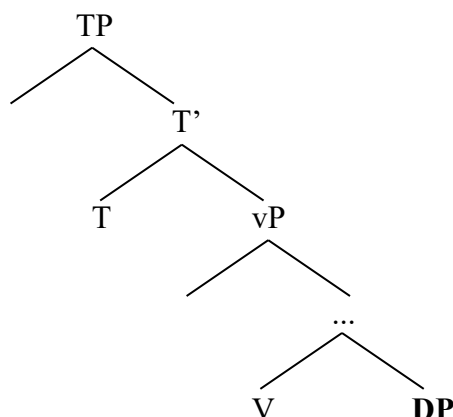
⁷ Kimura (2022a, *to appear*) and Kimura and Wakabayashi (2024) argue that an “obligatory” scrambling analysis is adopted at the intermediate stage, which is then followed by a target-like *wh*-movement analysis.

⁸ The discrepancy between the ease of resetting the head parameter and the difficulty in selecting $u\phi$ -features may be attributed to the fact that the former, belonging to the PF wing of grammar (Chomsky et al., 2019), is easily detectable in the input, whereas the latter is not directly observable, presenting a poverty-of-the-stimulus problem or being subject to the critical period effect (Hawkins & Hattori, 2006; Tsimpli & Dimitrakopoulou, 2007). The undetectability of $u\phi$ -features can also result in the absence of parsing failure (cf. Schwartz & Sprouse, 1996), thereby preventing access to UG at early stages of development. UG may function at later stages to facilitate uF acquisition (Kimura, 2022a, *to appear*; Kimura & Wakabayashi, 2024), but we will not explore the UG-access issue further here. We are grateful to the reviewer for bringing this important issue to our attention.

(16) a. *with scrambling*



b. *without scrambling*



In summary, elementary-level JLEs' interlanguage provides insight into the underlying syntactic structure of unergatives/unaccusatives. Given the above arguments, elementary-level JLEs may adopt optional scrambling for subjects. Additionally, given that head-directions are reset early, they may develop a grammar with English-headedness but optional subject scrambling. This leads to possible derivations for unaccusatives (17) and unergatives (18) as follows:

- | | | |
|---------|--|---------------------------------|
| (17) a. | [_{TP} ____ [_{VP} arrived the train]] | head-initial/without scrambling |
| b. | [_{TP} the train [_{VP} arrived ____]] | head-initial/with scrambling |
| (18) a. | [_{TP} ____ [_{VP} the boy run]] | head-initial/without scrambling |
| b. | [_{TP} the boy [_{VP} ____ run]] | head-initial/with scrambling |

Nevertheless, the subject-less configuration in (17a) is unlikely to be accepted by learners for independent reasons. Intermediate-level JLEs rarely drop subjects in L2 English (e.g., Wakabayashi & Negishi, 2003), and (17a) would likely be rejected. However, Kuribara (2004) found that intermediate-level JLEs accepted configurations where non-subject phrases, like adverbial topics, occupy the pre-verbal position (e.g., *Last Christmas had a wonderful party*). Kuribara suggests that JLEs have not acquired subject syntax but require the pre-verbal position to be filled by an overt phrase. Thus, we should modify the configurations in (17) to include an overt phrase in the pre-verbal position as follows:

- | | | |
|---------|--|---------------------------|
| (19) a. | *[When did [_{TP} ____ [_{VP} arrive the train]]]? | <i>without scrambling</i> |
| b. | [When did [_{TP} the train [_{VP} arrive ____]]]? | <i>with scrambling</i> |

With this modification, the sentences in (19) exhibit the following features: (i) headedness is reset to [head-initial], (ii) the subject is optionally scrambled, and (iii) the pre-verbal position is filled by an overt phrase. Thus, elementary-level JLEs may produce the sentences in (19), based on previous findings on Japanese-English interlanguage.

The crucial focus of this study is the unergative-unaccusative distinction, and our analysis of probable sentences in elementary-level JLEs' interlanguage should consider this distinction. Let us now examine the following two illicit configurations:

- (20) a. *[When did [TP ____ [VP arrive the train]]]? *unaccusative*
 b. *[Where did [TP ____ [VP dance the man]]]? *unergative*

These configurations would be acceptable only with unaccusative verbs, where the DP_{THEME} remains in Comp-VP, but not with unergative verbs, where the DP_{AGENT} cannot occupy that position if the grammar is sensitive to deep unaccusativity. Thus, elementary JLEs would accept (20a) but reject (20b). Moreover, if scrambling optionally applies, the configurations in (21a) would be as equally acceptable to (20a) for unaccusatives, while (21b) would be significantly more acceptable than (20b) for unergatives.

- (21) a. [When did [TP the train [VP arrive ____]]]? *unaccusative*
 b. [Where did [TP the man [VP ____ dance]]]? *unergative*

In sum, implications from the previous research suggest that elementary-level JLEs would generate an interlanguage where subjects can stay in situ, whereas the headedness is correctly reset to the English one. Therefore, this probable grammar would reveal the base-generated position of the subject, namely Spec-VP or Comp-VP.

5. Present Study

5.1 Research Question and Predictions

Against the background discussed above, this study seeks to answer the following research question (RQ):

RQ: Does interlanguage at early stages of development show sensitivity to the unergative-unaccusative distinction?

It has been suggested that L2 grammars at early stages consist of formulaic elements of

lexical categories (e.g., Klein & Purdue, 1997; Myles, 2004) and that L2 learners primarily rely on surface linear sequencing strategies (e.g., Meisel, 1991, 1997). Within this context, it is essential to investigate whether interlanguage at early stages of development is shaped by structural principles, including the unergative-unaccusative distinction.

5.2 Methodology

5.2.1 Participants

Thirty native speakers of English (NSEs) and 55 JLEs participated in this study. The NSEs, living in the U.S. at the time of testing, were recruited via Prolific.⁹ The JLEs, aged 18 to 20 and university students in Japan, were recruited from the authors' English classes and had no background in linguistics or related subjects. Both groups completed the task on the same platform (Google Forms).

The JLEs' English proficiency was assessed with the Oxford Quick Placement Test (University of Cambridge Local Examinations Syndicate, 2001) on a separate day before the main experiment. According to the test criteria, 28 were elementary-level, and 27 were intermediate-level. Detailed information is summarized in Table 1 below. (The test's maximum score is 60.)

Table 1

Summary of placement test scores

	<i>Range</i>	<i>Mean</i>	<i>SD</i>
elementary	13(14–27)	18.5	4.49
intermediate	12(30–42)	37.0	2.76

5.2.2 Task and Materials

We conducted an untimed acceptability judgment task on a 5-point scale. A sample item is given below:

⁹ Studies have shown that crowdsourcing services yield results comparable in quality to laboratory experiments (e.g., Sprouse, 2011).

Figure 1

A sample of task materials

Where did arrive the train? *						
	1	2	3	4	5	
Completely Unacceptable	☐	☐	☐	☐	☐	Completely Acceptable

The main experimental materials contained either unaccusative (*arrive, happen, die, fall, melt, freeze, close, appear*) or unergative verbs (*swim, run, walk, dance, sing, cry, laugh, shout*) selected from Levin and Rappaport-Hovav (1995). Given the discussion above, we put an adverbial *wh*-phrase (i.e., *when* or *where*) at the beginning of each sentence.¹⁰ Grammatical sentences contain licit sequences, namely the *Wh-Aux-DP-V* order (22a,c), whereas ungrammatical sentences contain illicit ones, namely the **Wh-Aux-V-DP* order (22b,d):

- | | | |
|---------|--------------------------------|-----------------------------------|
| (22) a. | Where did the train arrive? | <i>unaccusative/grammatical</i> |
| b. | *When did happen the accident? | <i>unaccusative/ungrammatical</i> |
| c. | Where did the man walk? | <i>unergative/grammatical</i> |
| d. | *When did run the man? | <i>unergative/ungrammatical</i> |

The number of the uses of *when* and *where* was equal across types and conditions. Eight sets of the main test stimuli were created and divided into two stimulus lists. Thus, each type contained four tokens in each list (16 in total). Thirty-six fillers, used in both lists, were also included, and the total number of stimuli that each participant saw was 52. Stimuli were manually pseudo-randomized (Ionin, 2012).

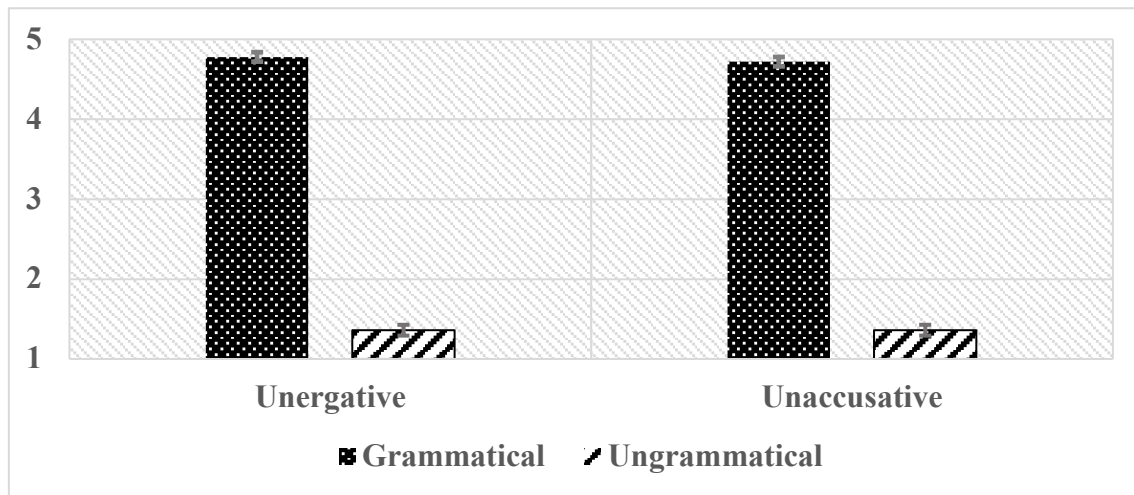
¹⁰ Alternative sentence-initial elements like temporal PPs (e.g., *At 5 pm*) or locative PPs (e.g., *On the stage*) pose other problems. Temporal PPs typically require topicalization with special phonology and a pause, often marked by a comma (e.g., *At 5 pm, walked the man./At 5 pm, arrived the train*), which might lead L2ers to consider that an overt pre-verbal phrase is missing, causing them to reject the sentence. Locative PPs create a grammatical construction called locative inversion, where the locative PP is fronted followed by a verb (e.g., *On the stage appeared a man* (Bresnan, 1994)). Thus, using locative PPs would test knowledge of a different construction. Considering these, we chose *wh*-questions as test items to minimize potential confounding factors.

6. Results

Looking at the mean acceptability scores, NSEs showed the expected patterns of responses, distinguishing grammatical and ungrammatical sentences (Figure 1).

Figure 2

Mean acceptability scores (NSEs)



Note. Error bars stand for standard errors of the mean (SE).

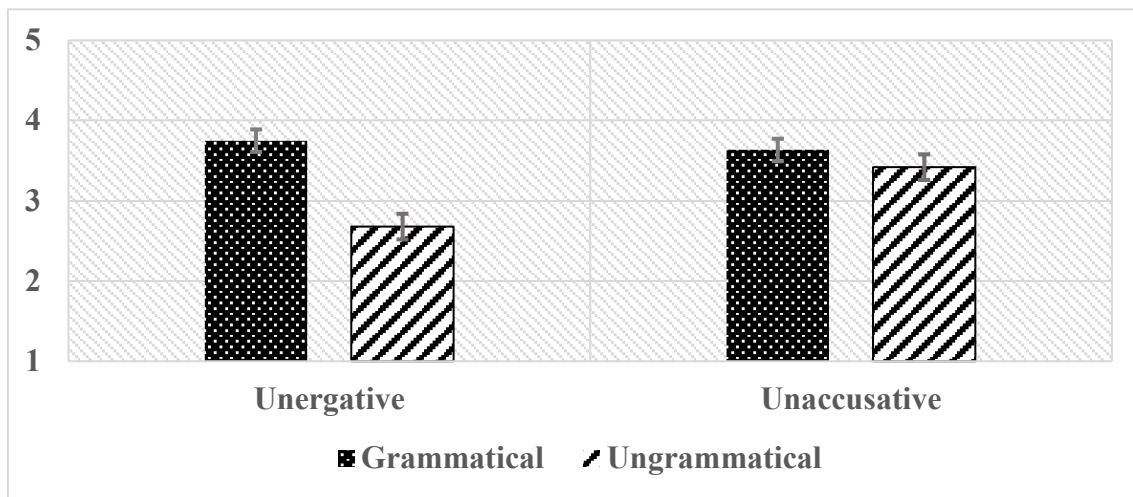
To estimate the effects of types and grammaticality on the judgments, response data were submitted to linear mixed-effects model analyses in *R* (Baayen et al., 2008). Prior to the analysis, fixed effects were dummy-coded: unaccusative was coded as ‘0’ and unergative as ‘1’, and ungrammatical was coded as ‘0’ and grammatical as ‘1’, with the coded values centered. The maximum model included verb types, grammaticality, and their interaction terms as fixed effects, and random slopes and intercepts for participants and items (maximum model: $\text{Response} \sim \text{Type} * \text{Grammaticality} + (1 + \text{Type} * \text{Grammaticality} | \text{Participants}) + (1 + \text{Type} * \text{Grammaticality} | \text{Item})$). The best-fitted model was selected for each group using a *step()* function. The models were implemented in *R* with the *lme4* package (Bates et al., 2020), and *p*-values were obtained by using an *lmerTest* package (Kuznetsova et al., 2017).

Regarding NSEs, the results of statistical estimates (final model: $\text{Response} \sim \text{type} * \text{Grammaticality} + (1 + \text{Grammaticality} | \text{Participants})$) suggest that only the main effect of grammaticality ($\beta = 3.39$, $SE = 0.09$, $t = 37.98$, $p < 0.001$), but not that of type ($\beta = 0.03$, $SE = 0.06$, $t = 0.56$, $p = 0.58$) or their interaction ($\beta = 0.07$, $SE = 0.11$, $t = 0.62$, $p = 0.54$), was significant, as expected by the theory.

Elementary JLEs showed greatly different behaviors from NSEs (Figure 3). In particular, a remarkable distinction was made between ungrammatical unergative sentences and the others. The mean acceptability scores for un/grammatical unaccusatives and grammatical unergatives were close to each other. The result of statistical estimation (final model: Response~Type*Grammaticality+(1+Grammaticality|Participants)) showed that a main effect of type ($\beta=-0.31$, $SE=0.12$, $t=-2.62$, $p<0.01$) and grammaticality ($\beta=0.62$, $SE=0.25$, $t=2.48$, $p=0.02$) as well as their interaction ($\beta=0.85$, $SE=0.24$, $t=3.60$, $p<0.001$) were all significant.

Figure 3

Mean acceptability scores (elementary JLEs)

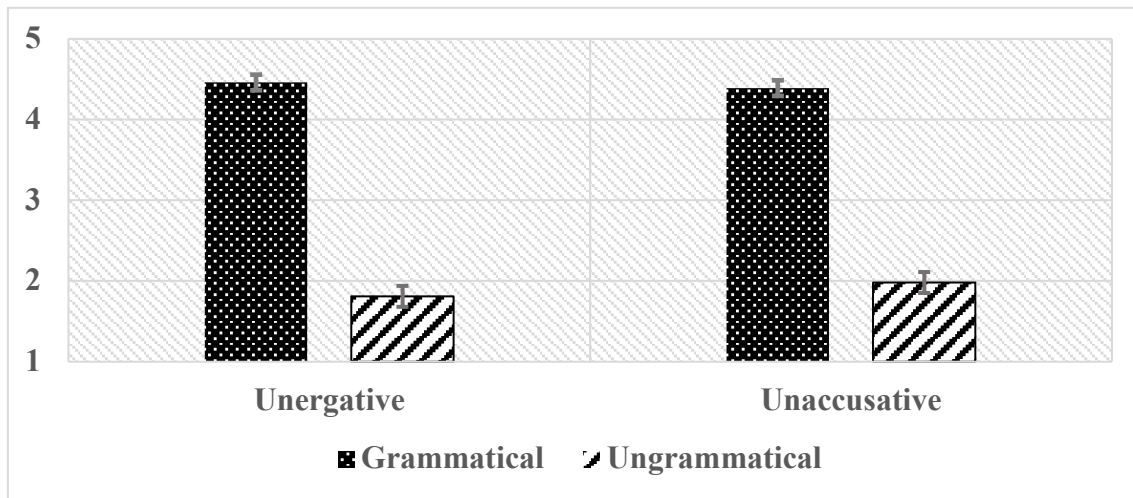


Furthermore, the results of pairwise comparisons (final model: Response~Pairwise_Type+(Grammaticality|Participants)) showed that responses for grammatical and ungrammatical unaccusatives did not differ significantly ($\beta=0.22$, $SE=0.28$, $t=0.79$, $p=0.44$), but those for ungrammatical unaccusatives and unergatives showed a significant difference ($\beta=-0.72$, $SE=0.17$, $t=-4.36$, $p<0.01$).

Next, responses from intermediate JLEs became closer to those from NSEs (Figure 4), as confirmed by statistics (final model: Response~Type*Grammaticality+(Type+Grammaticality|Participants): The main effect of grammaticality was significant ($\beta=2.53$, $SE=0.25$, $t=10.2$, $p<0.001$), whereas that of type and the interaction were not significant ($\beta=-0.05$, $SE=0.12$, $t=-0.39$, $p=0.70$ for type, $\beta=0.24$, $SE=0.21$, $t=1.2$, $p=0.23$ for the interaction).

Figure 4

Mean acceptability scores (intermediate JLEs)



To summarize, NSEs and intermediate JLEs significantly distinguished between grammatical and ungrammatical sentences regardless of verb type, as expected. In contrast, elementary JLEs showed a different pattern: They accepted ungrammatical unaccusative constructions to the same extent as grammatical ones but distinguished between verb types within ungrammatical conditions.

7. Discussion

Now, let us discuss the data and consider the syntactic knowledge of the unergative-unaccusative distinction in L2 grammars. First, let us consider the results from elementary JLEs, which are the main interest of this study. The experimental results showed that elementary-level JLEs accepted the ungrammatical **Wh-Aux-VP-DP* word order for the unaccusative verb construction to a similar extent as the grammatical counterpart. Furthermore, they distinguished it from the ungrammatical unergative verb construction ($M=2.68$), which was significantly lower-rated than the unaccusative counterpart ($M=3.42$).

We can explain the results as follows. In line with our hypothesis, let us assume i) that subject movement is optional via scrambling; and ii) the headedness parameter has been reset. Then, the acceptance of ungrammatical unaccusatives by elementary JLEs can be understood that in their interlanguage, the base-generated position for the DP_{THEME} is Comp-VP:

(23) ... [_{VP} ____ [_{VP} happen **the accident**]]

Moreover, the fact that the erroneous unergatives were significantly low-rated in comparison to the others suggests that in their interlanguage, the base-generated position for the DP_{AGENT} is Spec-VP but not Comp-VP, consistent with the structural unergative-unaccusative distinction:

(24) ... [_{VP} **the boy** [_{VP} run ____]]

The contrast above suggests that the construction of interlanguages is guided by the unergative-unaccusative distinction derived from UG. Instead, if L2 grammars relied on a linear-based system (e.g., Meisel, 1991, 1997), the distinction would be too hard or impossible to represent because, as noted, neither English nor Japanese show unaccusativity on the surface. Moreover, the reason that L2ers adopt scrambling may be associated with their effort to produce a linear order that aligns with the L2 input (i.e., the consistent presence of a subject in the subject position in English), as Kimura (2022a) and Miyamoto and Iijima (2003) discuss in the context of the acquisition of *wh*-questions.

Regarding intermediate JLEs, they behaved target-likely, distinguishing between grammatical and ungrammatical sentences. Furthermore, the discrepancy between ungrammatical unergatives and unaccusatives disappeared. These results suggest that their interlanguage moves toward the target-like representation, where obligatory subject movement obscures the difference between unergatives and unaccusatives, which is observed at the underlying structure level.

8. Conclusion

We investigated the interlanguage of elementary/intermediate JLEs concerning the syntax of unergatives/unaccusatives. The structural difference between these verb types is obscured in the surface strings of English and Japanese. Syntactic analyses suggest this is due to obligatory subject movement in English and head-finalness in Japanese. Based on the previous literature and theory, we predicted that elementary-level JLEs would develop an interlanguage with non-obligatory subject scrambling and a [head-first] setting, and we designed probable erroneous sentences accordingly. The experimental results suggest they constructed such a grammar, with their interlanguage constrained by the structural unergative/unaccusative distinction embodied in UG. Thus, we conclude that L2 grammar is governed by structural principles, and the developing Japanese-

English interlanguage provides new evidence for unergativity/unaccusativity in L2 grammar.

Competing Interests Declaration

There are no competing interests to declare.

Acknowledgements

We sincerely thank the three anonymous *LAB* reviewers and the editor for their invaluable comments and suggestions. We also appreciate the audience at BUCLD 49 for their insightful feedback. This work was partially supported by JSPS KAKENHI *Grant* Numbers JP24K16042 and JP24K16092.

Data availability statement

The raw data and analysis codes are available in the OSF link provided below (https://osf.io/rdpq9/?view_only=eb3921a5b44e4ac8b37cd3808decd955).

References

- Alexiadou, A., Anagnostopoulou, E., & Schäfer, F. (2015) *External arguments in transitivity alternations: A layering approach*. Oxford: Oxford University Press.
- Baayen, R. H., Davidson, D. J., & Bates, D. M. (2008) Mixed-effects modeling with crossed random effects for subjects and items. *Journal of Memory and Language* 59(4), 390–412.
- Baker, M. (1988) *Incorporation: A theory of grammatical function changing*. Chicago University Press.
- Baker, M., Johnson, K., & Roberts, I. (1989). Passive arguments raised. *Linguistic Inquiry* 20(2), 219–251.
- Balcom, P. (1997) Why is this happened? Passive morphology and unaccusativity. *Second Language Research* 13(1), 1–9.
- Bates, D., Maechler, M., Bolker, B., Walker, S., Christensen, R., Singmann, H., Dai, B., Scheipl, F., Grothendieck, G., & Green, P. (2020) Package ‘lme4’ (Version 1.1-26)
- Bresnan, J. (1994). Locative inversion and the architecture of universal grammar. *Language* 70(1), 72–131.
- Burzio, L. (1986) *Italian Syntax: A Government-binding Approach*. Reidel.
- Carrier, J., & Randall, J. H. (1992) The argument structure and syntactic structure of resultatives. *Linguistic Inquiry* 23(2), 173–234.

- Chomsky, N. (2001) Derivation by phase. In M. Kenstowicz (Ed.) *Ken Hale: a Life in Language* (pp.1–52). MIT Press.
- Collins, C. (2005) A smuggling approach to the passive in English. *Syntax* 8(2), 81–120.
- Epstein, S.D., Kitahara, H., & Seely, T.D. (2016) Phase Cancellation by external pairmerge of heads. *The Linguistic Review* 33, 87–102.
- Fukui, N. (1995) *Theory of projection in syntax*. CSLI Publications.
- Hawkins, R., & Hattori, H. (2006) Interpretation of English multiple wh-question by Japanese speakers: a missing uninterpretable feature account. *Second Language Research* 22(3), 269–301.
- Hawkins, R., & Liszka, S. (2003) Locating the source of defective past tense marking in advanced L2 English speakers. In R. van Hout, A. Hulk, F. Kuiken, & R. Towell (Eds.) *The interface between syntax and lexicon in second language acquisition* (pp. 21–44). John Benjamins.
- Haznedar, B. (1997) *Child second language acquisition of English: A longitudinal case study of a Turkish-speaking child* [Unpublished doctoral dissertation]. University of Durham.
- Haznedar, B., & Schwartz, B. D. (1997) Are there optional infinitives in child L2 acquisition? In E. Hughes, M. Hughes, & A. Greenhill (Eds.) *Proceedings of BUCLD 21* (pp.257–68). Cascadilla Press.
- Hirakawa, M. (2001) L2 acquisition of Japanese unaccusative verbs. *Studies in Second Language Acquisition* 23(2), 221–245.
- Hirakawa, M. (2003) *Unaccusativity in second language Japanese and English*. Kasugabe, Japan: Hituzi Syobo Publishing LTD.
- Hirakawa, M. (2006) ‘Passive’ unaccusative errors in L2 English revisited. In R. Slabakova, S. A. Montrul, & P. Prevost (Eds.) *Inquiries in linguistic development* (pp.17–39). John Benjamins.
- Hoekstra, T. (1988) Small clause results. *Lingua* 74(2-3), 101–139.
- Ionin, T. (2012) Formal theory-based methodologies. In A. Mackey, & S. M. Gass (Eds.) *Research methods in second language acquisition: A practical guide* (pp.30–52). Blackwell.
- Jaeggli, O. (1986) Passive. *Linguistic Inquiry* 17(4), 587–622.
- Ju, M. K. (2000) Overpassivization errors by second language learners: The effect of conceptualizable agents in discourse. *Studies in Second Language Acquisition* 22(1), 85–111.
- Kageyama, T. (1993) *Bunpoo to gokeisei* [Grammar and word formation]. Hitsuji Shobo.
- Kageyama, T. (1996) *Dooshi imi-ron* [Semantics of verbs]. Kuroshio Shuppan.

- Kimura, T. (2022a) Feature Selection, Feature Reassembly, and the Role of Universal Grammar: The Acquisition of *Wh*-questions by Japanese and Chinese Learners of English. [Unpublished doctoral dissertation]. Chuo University.
- Kimura, T. (2022b) ‘Control’-ed raising: Misanalyses of infinitival clause structures by L2 learners. *Journal of the Institute of Cultural Science* 103, 37–69.
- Kimura, T. (to appear) ‘Strong’ weak-island effects in Interlanguage: Arguments from D-linking. *Proceedings of BUCLD 49*.
- Kimura, T., & Wakabayashi, S. (2024) UG-as-Guide in selection and reassembly of an uninterpretable feature in L2 acquisition of *wh*-questions: Evidence from islands and scope. In M. Velnić, A. Dahl, & K. Listhaug (Eds.) *Current Perspectives on Generative SLA—Representations, Processing and Development* (pp.318–350). John Benjamins.
- Kishimoto, H. (1996) Split intransitivity in Japanese and the Unaccusative Hypothesis. *Language* 72(2), 248–286.
- Kuribara, C. (2004) Misanalysis of subjects in Japanese-English interlanguage. *Second Language* 3, 69-95.
- Kuznetsova, A., Brockhoff, P. A., & Christensen, R. H. B. (2017) lmerTest package: Tests in linear mixed effects models. *Journal of Statistical Software* 82(13), 1–26.
- Lardiere, D. (2009) Some thoughts on a contrastive analysis of features in second language acquisition. *Second Language Research* 25(2), 173–227.
- Leal, T ., & Shimanskaya, E. (2024) The power of paradox in bilingualism. *Linguistic Approaches to Bilingualism* 14(1), 85–89.
- Levin, B., & Rappaport-Hovav, M. (1995) *Unaccusativity: At the syntax-lexical semantics interface*. MIT Press.
- Lohndal, T., & Putnam, M. (2024) The importance of features and exponents. *Linguistic Approaches to Bilingualism* 14(1), 1–36.
- Meisel, J. (1991) Principles of Universal Grammar and strategies of language learning: some similarities and differences between first and second language acquisition. In Eubank L (ed) *Point counterpoint: Universal Grammar in the second language* (pp.231–76). John Benjamins.
- Meisel, J. (1997) The acquisition of the syntax of negation in French and German: contrasting first and second language acquisition. *Second Language Research* 13(3), 227–63.
- Miyamoto, Y., & Iijima, Y. (2003) On the existence of scrambling in the grammar of Japanese elementary EFL learners. In S. Foster-Cohen, & S. P. Doehler (Eds.) *Eurosla Yearbook* 3(1) (pp. 7–27). John Benjamins.

- Nakayama, M., & Yoshimura, N. (2019) On the intervention effects and subjects in the interpretation of raising constructions by Japanese learners of English. Talk at Tohoku University, Sendai, Japan. July 5th.
- Oshita, H. (2000) What is happened may not be what appears to be happening: A corpus study of 'passive' unaccusatives in L2 English. *Second Language Research* 16(4), 293–324.
- Park, K.-S., & Lakshmanan, U. (2007) The unaccusative-unergative distinction in resultatives: Evidence from Korean L2 learners of English. In A. Belikova, L. Meroni, & Umeda, M. (Eds.) *Proceedings of GALANA 2* (pp.328–338). Cascadilla Proceedings Project.
- Perlmutter, D. (1978) Impersonal passives and the Unaccusative Hypothesis. *Proceedings of the 4th Annual Meeting of the Berkeley Linguistics Society*, 157–189.
- Progovac, L. (2015) The absolutive basis of middles and the status of vP and UTAH. *FASL* 23, 242–261.
- Rappaport-Hovav, M., & Levin, B. (2001) An event structure account of English resultatives. *Language* 77, 766–797.
- Saito, M. (2003) A derivational approach to the interpretation of scrambling chains. *Lingua* 113: 481–518.
- Schwartz, B. D., & Sprouse, R. A. (1996) L2 cognitive states and the Full Transfer/Full Access model. *Second Language Research* 12(1), 40–72.
- Sprouse, J. (2011) A validation of Amazon Mechanical Turk for the collection of acceptability judgments in linguistic theory. *Behavior Research Methods* 43(1), 155–167.
- Tsimpli, I. M., & Dimitrakopoulou, M. (2007) The Interpretability Hypothesis: Evidence from *wh*-interrogatives in second language acquisition. *Second Language Research* 23(2), 215–242.
- University of Cambridge Local Examinations Syndicate. (2001) Quick Placement Test. Oxford: Oxford University Press.
- Wakabayashi, S., & Negishi, R. (2003) Asymmetry of subjects and objects in Japanese speakers' L2 English. *Second Language* 2, 53–73.
- Wechsler, S. (1997) Resultative predicates and control. *Proceedings of the Texas Linguistic Forum* 38, University of Texas, Austin, pp. 307–321.
- White, L. (2003) *Second language acquisition and Universal Grammar*. Cambridge University Press.
- Yatsushiro, K. (1999) Case Licensing and VP Structure. [Unpublished doctoral dissertation]. University of Connecticut.

- Yusa, N. (2003) 'Passive' unaccusatives in L2 acquisition. In P. Clancy (Ed.) *Japanese/Korean Linguistics 11* (pp.246–259). CLSI.
- Zobl, H. (1989) Canonical typological structures and ergativity in English L2 acquisition. In S. Gass, & J. Schachter (Eds.) *Linguistic perspectives on second language acquisition* (pp.203–21). Cambridge University Press.